Lecture 07 – Reinforcement Learning

Stat 5376: Statistical Machine Learning

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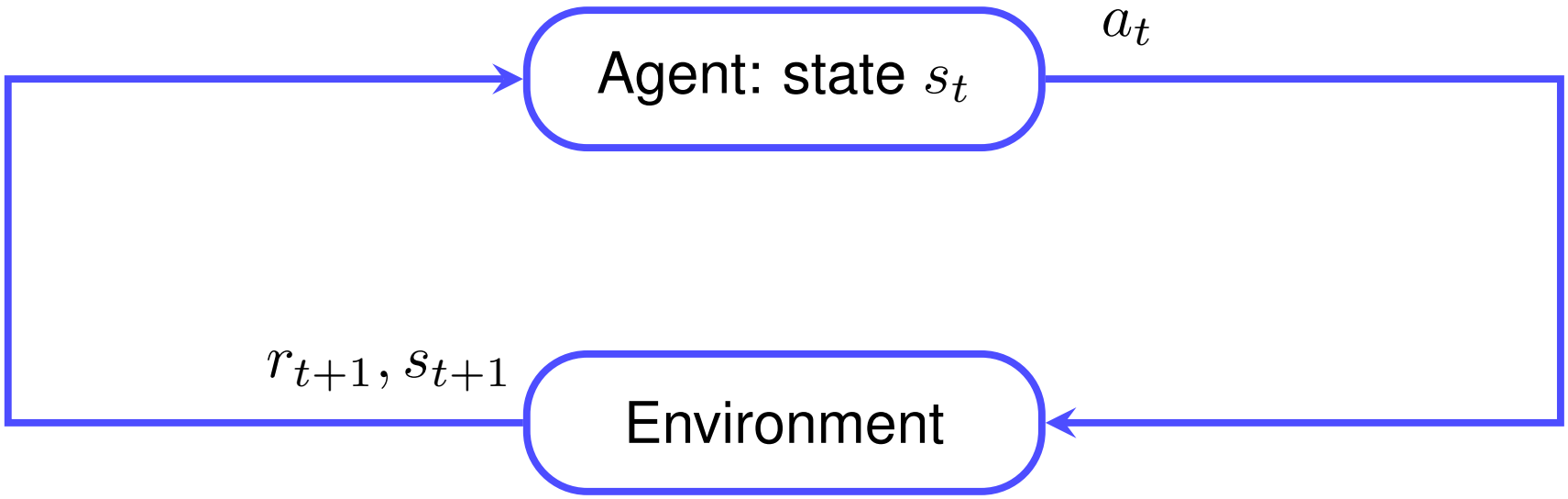
Markov Decision Process

- Model:



  - Current state: $s_t \in \mathcal{S}$
  - Action: $a_t \in \mathcal{A}$
  - Reward: $r_{t+1} \in \mathbb{R}$
  - Next state: $s_{t+1} \in \mathcal{S}$

Example

- Frozen lake: Find a path from S (start) to G (goal)

S	F	F	F
F	H	F	H
F	F	F	H
H	F	F	G

  - State set: $\mathcal{S} = \{1, 2, \ldots, 16\}$
  - Action set: $\mathcal{A} = \{\text{up, down, left, right}\}$
  - Reward: $r_{t+1} = 0$ if $s_{t+1} \neq 16$ and $r_{t+1} = 1$ if $s_{t+1} = 16$
- You also can define $r_{t+1} = -100$ if $s_{t+1} = 6, 8, 12, 13$ (H).

Mathematical Model

- Transition probability: $p(s' = s_{t+1} | s = s_t, a = a_t)$
  - Probaiblity of next state is s_{t+1} given current state is s_t and action is a_t
  - In some situations, next state is deterministic given current state and action
  - So $s' = s_{t+1}$ if $s = s_t$ and $a = a_t$ or $p(s' = s_{t+1} | s = s_t, a = a_t) = 1$
- Reward: $r_{t+1} = r(s_t, a_t, s_{t+1})$
  - Reward is a random variable given current state s_t, action a_t, and next state s_{t+1}
  - In many situations, reward only depends on the new state
  - So it becomes $r_{t+1} = r(s_{t+1})$
- (s_t, a_t, s_{t+1}): commonly used block

Example

- Vanilla frozen lake:

S	F	F	F
F	H	F	H
F	F	F	H
H	F	F	G

- Next state is derministic given current state and action:
  - $p(s' = 2 | s = 1, a = \text{right}) = 1$
  - $p(s' = 5 | s = 1, a = \text{down}) = 1$
- Reward only depends on new state:
  - $r(16) = 1, r(3) = 0$

Example

- Slippery frozen lake:

S	F	F	F
F	H	F	H
F	F	F	H
H	F	F	G

- The lake is slippery, so that the player may move perpendicular to the intended direction sometimes.
- Next state is random given current state and action:
  - $p(s' = 2|s = 1, a = \text{right}) = 0.5$, $p(s' = 5|s = 1, a = \text{right}) = 0.5$
  - $p(s' = 11|s = 7, a = \text{down}) = 1/3$, $p(s' = 6|s = 7, a = \text{down}) = 1/3$, $p(s' = 8|s = 7, a = \text{down}) = 1/3$

Policy

- A policy π is the agent's internal strategy on picking actions.
- Deterministic policies: maps state and action directly:

$$a_t = \pi(s_t).$$

- Stochastic policies: maps a probability of the action given a state:

$$\pi(a_t|s_t) = \text{ probability of choosing action } a_t \text{ given current state } s_t.$$

Example

- Frozen lake: Find a path from S (start) to G (goal)

S	F	F	F
F	H	F	H
F	F	F	H
H	F	F	G

- Detemrinistic policy:
  - $\pi(1) = \text{down}, \pi(5) = \text{down}, \pi(9) = \text{right}, \pi(10) = \text{right}, \pi(11) = \text{down}, \pi(15) = \text{right}$
  - For other state s, the policy $\pi(s) = \text{right}$.

Example

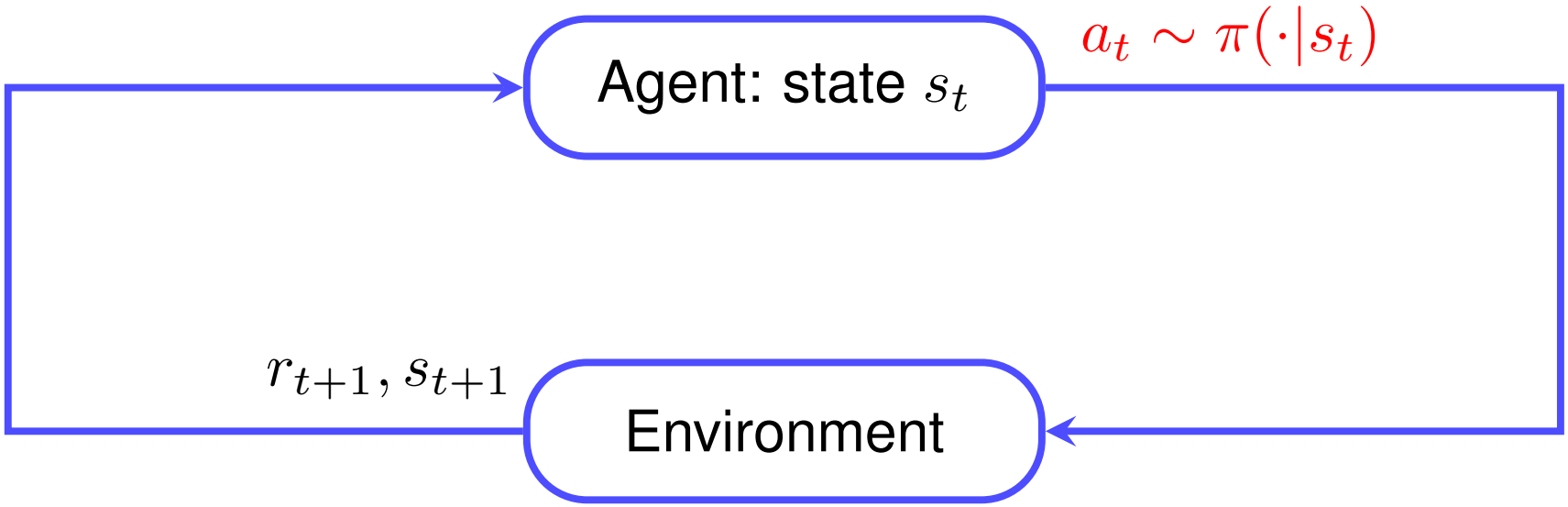
- Frozen lake: Find a path from S (start) to G (goal)

S	F	F	F
F	H	F	H
F	F	F	H
H	F	F	G

- Stochastic policy:
  - Uniform policy: $\pi(\text{up}|s) = \pi(\text{down}|s) = \pi(\text{right}|s) = \pi(\text{left}|s) = 1/4$

Markov Decision Process

- Model with a policy:



- Under the policy π, the agent will recieve rewards: $r_1, r_2, \ldots$.
- The total expected discounted reward is

$$\mathrm{E}_\pi \left\{ \sum_{k=0}^{\infty} \gamma^k r_{k+1} \right\} = \mathrm{E}_\pi \left\{ r_1 + \gamma r_2 + \gamma^2 r_3 + \ldots \right\}$$

  - Discounted factor: $\gamma \in (0, 1)$
  - The rewards are random and depend on π, so we use E_π

Example

- Frozen lake: Find a path from S (start) to G (goal)

S	F	F	F
F	H	F	H
F	F	F	H
H	F	F	G

- Policy 1:
  - $\pi(1) = \text{down}, \pi(5) = \text{down}, \pi(9) = \text{right}, \pi(10) = \text{right}, \pi(11) = \text{down}, \pi(15) = \text{right}$
  - Total reward: $\mathrm{E}_\pi \left\{ \sum_{k=0}^{\infty} \gamma^k r_{k+1} \right\} = \gamma^5$
- Policy 2:
  - $\pi(1) = \text{right}, \pi(2) = \text{right}, \pi(3) = \text{down}, \pi(7) = \text{down}, \pi(11) = \text{left}, \pi(10) = \text{down}, \pi(14) = \text{right}, \pi(15) = \text{right}$
  - Total reward: $\mathrm{E}_\pi \left\{ \sum_{k=0}^{\infty} \gamma^k r_{k+1} \right\} = \gamma^8$

Evaluation Functions

- Value function:

$$V^{\pi}(s) = \mathrm{E}_{\pi}\left\{\sum_{k=0}^{\infty}\gamma^{k}r_{k+1}\middle| s_0 = s\right\}$$

- Under policy π and with initial state s, the value function $V^{\pi}(s)$ is the total expected discounted reward.
- Value-Action function:

$$Q^{\pi}(s,a) = \mathrm{E}_{\pi}\left\{\sum_{k=0}^{\infty}\gamma^{k}r_{k+1}\middle| s_0 = s, a_0 = a\right\}$$

- Under policy π and with initial state s, inital action a, the value-action function $Q^{\pi}(s,a)$ is the total expected discounted reward.
- Simple fact: $V^{\pi}(s) = \sum_a \pi(a|s)Q^{\pi}(s,a)$
  - Proof by the definition of conditional probaiblity and expectation
  - If π is derministic, then $V^{\pi}(s) = Q^{\pi}(s,\pi(s))$

Example

- Frozen lake: Find a path from S (start) to G (goal)

S	F	F	F
F	H	F	H
F	F	F	H
H	F	F	G

- Policy:
  - $\pi(1) = \text{down}, \pi(5) = \text{down}, \pi(9) = \text{right}, \pi(10) = \text{right}, \pi(11) = \text{down}, \pi(15) = \text{right}$
  - $V^\pi(1) = \gamma^5, V^\pi(11) = \gamma$
  - $Q^\pi(1, \text{down}) = Q^\pi(1, \pi(1)) = \gamma^5$, $Q^\pi(11, \text{left}) = \gamma^3$

Bellman Equation

- By the definition of Q^π, we have

$$
\begin{aligned}
Q^\pi(s,a) &= \mathrm{E}_\pi\left\{\sum_{k=0}^{\infty}\gamma^k r_{k+1}\middle| s_0 = s, a_0 = a\right\} \\
&= \mathrm{E}_\pi\left\{r_1 + \sum_{k=1}^{\infty}\gamma^k r_{k+1}\middle| s_0 = s, a_0 = a\right\} \\
&= \mathrm{E}_\pi\left\{r_1\middle| s_0 = s, a_0 = a\right\} + \mathrm{E}_\pi\left\{\sum_{k=1}^{\infty}\gamma^k r_{k+1}\middle| s_0 = s, a_0 = a\right\}
\end{aligned}
$$

- First item is easy:

$$
\begin{aligned}
\mathrm{E}_\pi\left\{r_1\middle| s_0 = s, a_0 = a\right\} &= \mathrm{E}_\pi\left\{r(s_0, a_0, s_1)\middle| s_0 = s, a_0 = a\right\} \\
&= \mathrm{E}_\pi\left\{r(s, a, s_1)\middle| s_0 = s, a_0 = a\right\} \\
&= \sum_{s'} p(s'|s,a) r(s,a,s')
\end{aligned}
$$

Bellman Equation

- Second iterm is tricky:

$$
\begin{aligned}
&\mathrm{E}_\pi\left\{\sum_{k=1}^{\infty}\gamma^k r_{k+1}\middle| s_0=s, a_0=a, s_1=s'\right\}\\
&=\gamma\mathrm{E}_\pi\left\{\sum_{k=1}^{\infty}\gamma^{k-1} r_{k+1}\middle| s_0=s, a_0=a, s_1=s'\right\}\\
&=\gamma\mathrm{E}_\pi\left\{\sum_{t=0}^{\infty}\gamma^{t} r_{t+2}\middle| s_0=s, a_0=a, s_1=s'\right\}\\
&=\gamma\mathrm{E}_\pi\left\{\sum_{t=0}^{\infty}\gamma^{t} r_{t+2}\middle| s_1=s'\right\}=\gamma V^\pi(s')
\end{aligned}
$$

- Hence, we have

$$
\mathrm{E}_\pi\left\{\sum_{k=1}^{\infty}\gamma^k r_{k+1}\middle| s_0=s, a_0=a\right\}=\sum_{s'}p(s'|s,a)\gamma V^\pi(s')
$$

Bellman Equation

- Bellman equation I:

$$Q^{\pi}(s,a) = \sum_{s'} p(s'|s,a)\left(r(s,a,s') + \gamma V^{\pi}(s')\right) \tag{1}$$

- Bellman equation II:

$$Q^{\pi}(s,a) = \sum_{s'} p(s'|s,a)\left(r(s,a,s') + \gamma \sum_{a'} \pi(a'|s') Q^{\pi}(s',a')\right) \tag{2}$$

  - Prove using $V^{\pi}(s) = \sum_a \pi(a|s) Q^{\pi}(s,a)$

Optimal Policy

- Optimal policy:

$$V^{\pi^*}(s) \geq V^{\pi}(s), \text{ for all policy } \pi \text{ and all initial state } s \ .$$

  - π^* simultaneously maximizes $V^{\pi}(s)$ for all s.
  - Opitmal policy is not unique; see the frozen lake exmaple.
- Define optimal value functions:

$$V^*(s) = \max_{\pi} V^{\pi}(s), \quad Q^*(s,a) = \max_{\pi} Q^{\pi}(s,a)$$

  - The definition implies that $V^*(s) = V^{\pi^*}(s)$
  - But we only get $Q^{\pi^*}(s,a) \leq Q^*(s,a)$ at this moment, and will show that they are the same later!

Greedy Policy and Policy Improvement Theorem

- Given a policy π and its value-action function $Q^\pi(s,a)$, we can define a greedy policy as follows:

$$\pi'(s) = \operatorname*{argmax}_a Q^\pi(s,a)$$

- Example:

Table: $Q^\pi(s,a)$

		Action		
		1	2	3
State	1	1.3	**5.2**	-3
	2	**4.3**	3.5	2.3
	3	5.1	3.2	**8.8**
	4	-0.3	**4.2**	2.3

$$\pi'(1) = 2,$$
$$\pi'(2) = 1,$$
$$\pi'(3) = 3,$$
$$\pi'(4) = 2.$$

- Policy Improvement Theorem: $V^{\pi'}(s) \geq V^\pi(s)$ for all states s
  - Our goal is to find a policy π to maximize the value function $V^\pi(s)$
  - Policy improvement theorem tells we can always using the greedy policy to improve the value function: $\pi \to Q^\pi \to \pi'$
  - The proof is omitted

Optimal Value Functions Equation

- Goal: $V^*(s) = \max_a Q^*(s,a)$
- By the definition of V^* and Q^{π^*}, we have

$$\begin{aligned} V^*(s) = V^{\pi^*}(s) &= \sum_a \pi^*(a|s) Q^{\pi^*}(s,a) \\ &\overset{\text{(i)}}{\leq} \max_a Q^{\pi^*}(s,a) \overset{\text{(i)}}{\leq} \max_a Q^*(s,a) \end{aligned} \tag{3}$$

  - (i) is due to the fact that average is smaller than maximum
  - (ii) holds by the definition of Q^*

Optimal Value Functions Equation

- Consider a class of policies, Π_a, that commit action a at periord 0 (i.e., $a_0 = a$) regardless the state
- So we have

$$
\begin{aligned}
Q^*(s,a) = \sup_{\pi} Q^{\pi}(s,a) &= \sup_{\pi \in \Pi_a} Q^{\pi}(s,a) \\
&= \sup_{\pi \in \Pi_a} \mathrm{E}_{\pi}\left\{ \sum_{k=0}^{\infty} \gamma^k r_{k+1} \middle| s_0 = s, a_0 = a \right\} \\
&= \sup_{\pi \in \Pi_a} \mathrm{E}_{\pi}\left\{ \sum_{k=0}^{\infty} \gamma^k r_{k+1} \middle| s_0 = s, a_0 = \pi(s) \right\} \\
&= \sup_{\pi \in \Pi_a} V^{\pi}(s) \leq \sup_{\pi} V^{\pi}(s) = V^*(s).
\end{aligned}
$$

- Taking maximum over a, we have

$$
\max_a Q^*(s,a) \leq V^*(s).
$$

Optimal Value Functions Equation

- Using the previous two slides, we have

$$V^{\pi^*}(s) = V^*(s) = \max_a Q^*(s,a)$$

- Using the above equation and (3), it holds that

$$V^*(s) = V^{\pi^*}(s) \leq \max_a Q^{\pi^*}(s,a) \leq \max_a Q^*(s,a) = V^*(s)$$

- Optimal value functions equation:

$$V^{\pi^*}(s) = V^*(s) = \max_a Q^*(s,a) = \max_a Q^{\pi^*}(s,a) \tag{4}$$

- At this moment, we do not have $Q^*(s,a) = Q^{\pi^*}(s,a)$

Bellman Opitmality Equation

- By the definition of Q^* and (1), we have

$$
\begin{aligned}
Q^*(s,a) &= \max_{\pi} Q^{\pi}(s,a) \\
&= \max_{\pi} \sum_{s'} p(s'|s,a)\left(r(s,a,s') + \gamma V^{\pi}(s')\right) \\
&= \sum_{s'} p(s'|s,a)\left(r(s,a,s') + \gamma \max_{\pi} V^{\pi}(s')\right) \\
&= \sum_{s'} p(s'|s,a)\left(r(s,a,s') + \gamma V^{\pi^*}(s')\right) \qquad (5)
\end{aligned}
$$

- Taking maximum over a and using (4), we have Bellman opitmality equation I:

$$
V^{\pi^*}(s) = \max_{a} \sum_{s'} p(s'|s,a)\left(r(s,a,s') + \gamma V^{\pi^*}(s')\right) \qquad (6)
$$

Bellman Opitmality Equation

- Using (5) and (4), we have Bellman opitmality equation II:

$$Q^*(s,a) = \sum_{s'} p(s'|s,a)\left(r(s,a,s') + \gamma \max_a Q^*(s',a)\right) \tag{7}$$

- By (2) and (4), we have

$$\begin{aligned} Q^{\pi^*}(s,a) &= \sum_{s'} p(s'|s,a)\left(r(s,a,s') + \gamma V^{\pi^*}(s')\right) \\ &= \sum_{s'} p(s'|s,a)\left(r(s,a,s') + \gamma \max_a Q^*(s',a)\right) \end{aligned}$$

- Combining the previous two equations, we conclude that

$$Q^*(s,a) = Q^{\pi^*}(s,a)$$

Summary

- Bellman equation:

$$Q^{\pi}(s,a) = \sum_{s'} p(s'|s,a)\left(r(s,a,s') + \gamma V^{\pi}(s')\right),$$

$$Q^{\pi}(s,a) = \sum_{s'} p(s'|s,a)\left(r(s,a,s') + \gamma \sum_{a'} \pi(a'|s') Q^{\pi}(s',a')\right).$$

- Optimal value functions equation:

$$Q^{\pi^*}(s,a) = Q^*(s,a),$$
$$V^{\pi^*}(s) = V^*(s) = \max_a Q^{\pi^*}(s,a) = \max_a Q^*(s,a).$$

- Bellman optimality equation:

$$V^{\pi^*}(s) = \max_a \sum_{s'} p(s'|s,a)\left(r(s,a,s') + \gamma V^{\pi^*}(s')\right),$$

$$Q^{\pi^*}(s,a) = \sum_{s'} p(s'|s,a)\left(r(s,a,s') + \gamma \max_a Q^{\pi^*}(s',a)\right).$$

Value Iteration

- Use Bellman equation to estimate Q^{π^*}:

$$Q^{\pi^*}(s,a) = \sum_{s'} p(s'|s,a)\left(r(s,a,s') + \gamma V^{\pi^*}(s')\right)$$

- Use Bellman opitmal equation to estimate V^{π^*}:

$$V^{\pi^*}(s) = \max_a \sum_{s'} p(s'|s,a)\left(r(s,a,s') + \gamma V^{\pi^*}(s')\right) = \max_a Q^{\pi^*}(s,a)$$

- Iteratively update Q^{π^*} and V^{π^*}

Value Iteration

Algorithm 1 Value Iteration

Require: Initialize $V(s) = 0$ for all $s \in \mathcal{S}$
Require: Choose an error tolerance $\epsilon > 0$ small
Require: Set initially $\Delta \leftarrow 1000000$ very large
  while $\Delta > \epsilon$ **do**
    for `all` s **do**
      for `all` a **do** # Bellman equation
        $Q(s, a) \leftarrow \sum_{s'} p(s'|s, a) \left(r(s, a, s') + \gamma V(s')\right)$
      end for
      $V_{\text{old}}(s) = V(s)$
      $V(s) \leftarrow \max_a Q(s, a)$ # Bellman optimality equation
    end for
    $\Delta \leftarrow \max_s |V_{\text{old}}(s) - V(s)|$
  end while
  Output policy $\pi^*(s) = \operatorname{argmax}_a Q(s, a)$

Policy Iteration

- Use Bellman equation to estimate Q^π:

$$Q^\pi(s,a) = \sum_{s'} p(s'|s,a)\left(r(s,a,s') + \gamma \sum_{a'} \pi(a'|s') Q^\pi(s',a')\right)$$

- Use policy improvement theorem to update π:

$$\pi' = \operatorname*{argmax}_a Q^\pi(s,a),$$
$$\pi = \pi'$$

- Iteratively use the previous two steps to update π

Policy Iteration

Algorithm 2 Policy Iteration

Require: Initialize policies $\pi(s) = 0$ for all $s \in \mathcal{S}$ and $\pi' \neq \pi$
Require: Initialize $Q(s, a) = 0$ for all $s \in \mathcal{S}, a \in \mathcal{A}$
Require: Choose an error tolerance $\epsilon > 0$ small

```
while π' ≠ π do
    Set initially Δ ← 1000000 very large
    while Δ > ε do # Bellman equation to estimate Q(s,a) = Q^π(s,a)
        for all (s,a) do
            Q_old(s,a) ← Q(s,a)
            Q(s,a) ← Σ_{s'} p(s'|s,a) (r(s,a,s') + γ Σ_{a'} π(a'|s') Q(s',a'))
        end for
        Δ ← max_{s,a} |Q_old(s,a) − Q(s,a)|
    end while
    for all s do
        π'(s) = π(s)
        π(s) = argmax_a Q(s,a) # policy improvement
    end for
end while
Output policy π(s)
```

Limitations of Value/Policy Iteration

- Need to know the transition probabilities $p(s'|s, a)$
- Need to know the complete information of the reward $r(s, a, s')$
- Are they reasonable?

Q-Learning

Algorithm 3 Q-Learning

Require: Initialize $Q(s, a) = 0$ for all s, a
Require: Some policy π_b (behavior policy)
Require: Learning rate α, number of epoches R
  for loop R times **do**
    Initial game and observe current state x
    while game is not over **do**
      Take action a based on s and behavior policy π_b
      Receive reward r and observe next state s'
      $Q(s, a) \leftarrow Q(s, a) + \alpha\big(r + \gamma \max_{a'} Q(s', a') - Q(s, a)\big)$
      $s \leftarrow s'$
    end while
  end for
  Output policy $\pi^*(s) = \operatorname{argmax}_a Q(s, a)$

Behavior Policy π_b

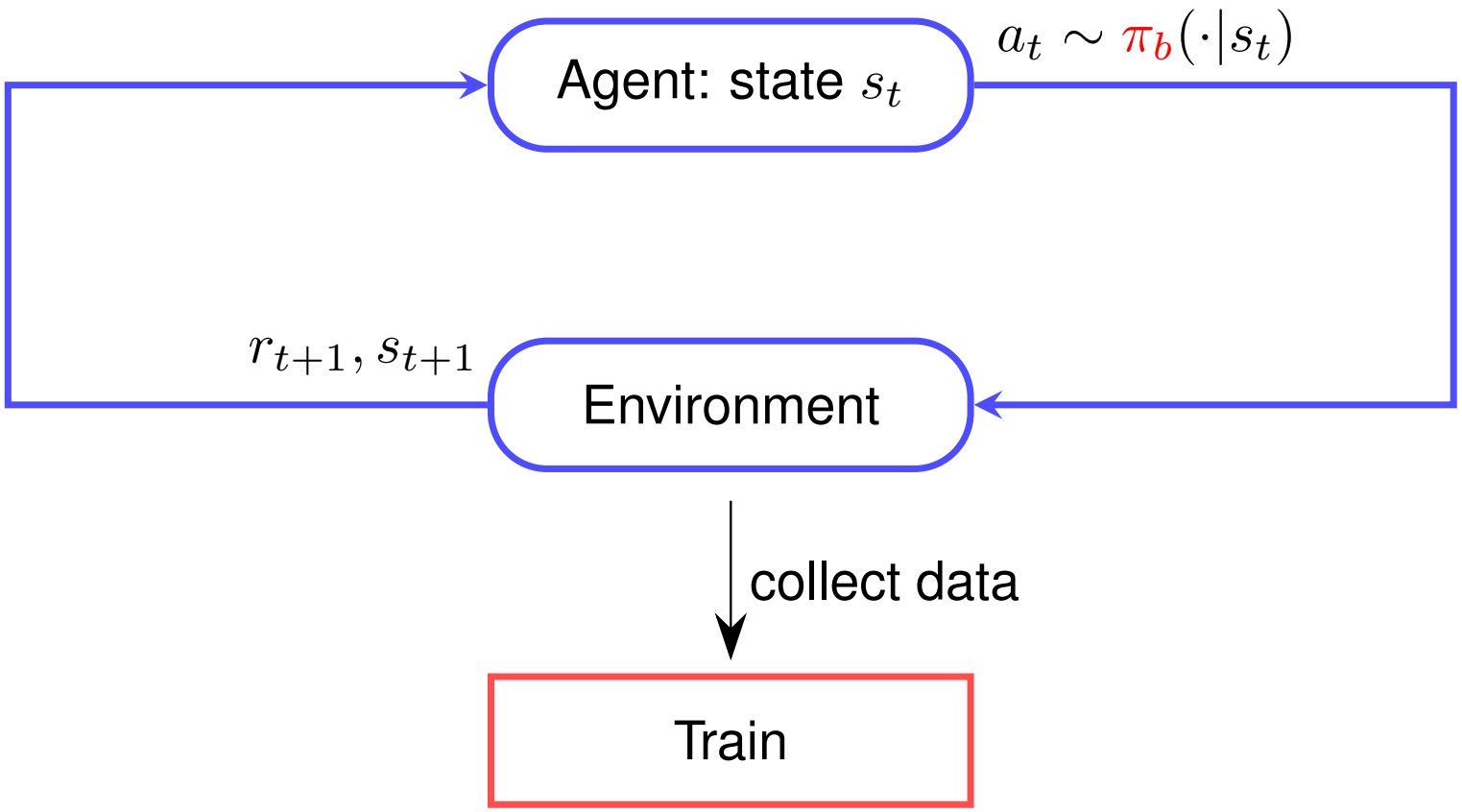
- Uniform Policy: Given any state s, the policy π_b has equal chance to choose each action.
- ϵ-greedy Policy: Given current Q-table, the optimal action given state s is $a_x^* = \mathrm{argmax}_a\, Q(s, a)$. The ϵ-greedy policy π_b given Q-table and current state s is

$$\begin{cases} a_x^* & \text{with probability } 1-\epsilon, \\ \text{randomly choose an action } a & \text{with probability } \epsilon. \end{cases}$$

- ϵ can be decreasing with the iterations.

Offline Reinforcement Learning

- Experiment is expensive, not everyone can interact with the envrioment.
- A person interact with the enviroment using a behavior policy π_b, and collect data: $(\text{current state}, \text{action}, \text{reward}, \text{next state})$
- Other people use the data to learn a policy.
- Work flow:



Offline Q-Learning

Algorithm 4 Offline Q-Learning

Require: Fixed dataset $\mathcal{D} = \{(s_t, a_t, r_t, s'_t)\}_{t=1}^{N}$, here $s'_t = s_{t+1}$
Require: Initialize $Q(s, a) = 0$ for all s, a
Require: Learning rate α, discount factor γ, number of epoches R
  for epoch $= 1, \ldots, R$ **do**
    for each transition $(s, a, r, s') \in \mathcal{D}$ **do**
      $Q(s, a) \leftarrow Q(s, a) + \alpha\Big(r + \gamma \max_{a'} Q(s', a') - Q(s, a)\Big)$
    end for
  end for
  Output policy $\pi^*(s) = \arg\max_a Q(s, a)$

Coding

- See instruction in class

More about Q-Learning

- Limitations of Q-table based methods
  - Infeasible when state space or action space is large, see Pong example
  - DQN: Use neural network $Q_\theta(s, a)$ to approximate Q-table

Policy Gradient Method

- Q-table based methods:
  - Find the optimal Q function: $Q^*(s, a)$
  - Find the greedy policy: $\pi^*(s) = \mathrm{argmax}_a \, Q^*(s, a)$
- Gradient based methods:
  - Parametrize the policy π_θ
  - Use gradient method updating θ to find the opitmal value of

$$J(\theta) = \sum_s \rho_0(s) V^{\pi_\theta}(s)$$

  - $\rho_0(s)$ is the probablity of inital state $s_0 = s$
  - In the frozen like exmaple, $\rho_0(1) = 1$ as it always starts at $s_0 = 1$.
- Softmax Policy:

$$\pi_\theta(a|s) = \frac{e^{\theta_{s,a}}}{\sum_{a'} e^{\theta_{s,a'}}} \text{ with } \theta = (\theta_{s,a}) \in \mathbb{R}^{|\mathcal{S}| \times |\mathcal{A}|}$$

Policy Gradient Method

- Suppose $s_0 \sim \rho_0$ and we have a trajectory

$$\tau = (s_0, a_0, r_0, s_1, a_1, r_1, \ldots) \text{ generated from policy } \pi_\theta$$

- We can find the distribution

$$\begin{aligned} p_\theta(\tau) &= \rho_0(s_0)\pi_\theta(a_0|s_0)p(s_1|s_0,a_0) \times \pi_\theta(a_1|s_1)p(s_2|s_1,a_2) \times \ldots \\ &= \rho_0(s_0)\prod_{t=0}^{\infty}\pi_\theta(a_t|s_t)p(s_{t+1}|s_t,a_t) \end{aligned}$$

- Hence the value functions is

$$J(\theta) = \sum_s \rho_0(s)V^{\pi_\theta}(s) = \sum_\tau p_\theta(\tau)\sum_{k=0}^{\infty}\gamma^k r_{k+1} = \sum_\tau p_\theta(\tau)R(\tau)$$

Policy Gradient Method

- Now we have

$$p_\theta(\tau) = \rho_0(s_0) \prod_{t=0}^{\infty} \pi_\theta(a_t|s_t) p(s_{t+1}|s_t, a_t),$$

$$J(\theta) = \sum_\tau p_\theta(\tau) R(\tau)$$

- Use the traick $\nabla p_\theta = p_\theta \nabla \log p_\theta$, we have

$$\begin{aligned}
\nabla J(\theta) &= \sum_\tau R(\tau) p_\theta(\tau) \nabla \log p_\theta(\tau) \\
&= \sum_\tau R(\tau) p_\theta(\tau) \sum_{t=0}^{\infty} \nabla \log \pi_\theta(a_t|s_t) \\
&= \mathrm{E}_{\pi_\theta} \left\{ R(\tau) \sum_{t=0}^{\infty} \nabla \log \pi_\theta(a_t|s_t) \right\}
\end{aligned}$$

Policy Gradient Method

- If $k < t$, we have

$$\mathrm{E}_{\pi_\theta}\left\{r_k \nabla \log \pi_\theta(a_t|s_t)\right\} = \mathrm{E}_{\pi_\theta}\left\{r(s_k, a_k, s_{k+1}) \nabla \log \pi_\theta(a_t|s_t)\right\} \overset{\text{(i)}}{=} 0$$

- To verify (i), we condition on $(s_0, a_0, r_0, s_1, \ldots, s_t)$. Therefore, it holds that

$$\begin{aligned}
&\mathrm{E}_{\pi_\theta}\left\{r(s_k, a_k, s_{k+1}) \nabla \log \pi_\theta(a_t|s_t) \middle| s_0, a_0, r_0, s_1, \ldots, s_t\right\} \\
&= r(s_k, a_k, s_{k+1}) \mathrm{E}_{\pi_\theta}\left\{\nabla \log \pi_\theta(a_t|s_t) \middle| s_0, a_0, r_0, s_1, \ldots, s_t\right\} \\
&= r(s_k, a_k, s_{k+1}) \mathrm{E}_{\pi_\theta}\left\{\nabla \log \pi_\theta(a_t|s_t) \middle| s_t\right\} \\
&= r(s_k, a_k, s_{k+1}) \sum_{a_t} \pi_\theta(a_t|s_t) \nabla \log \pi_\theta(a_t|s_t) \\
&= r(s_k, a_k, s_{k+1}) \sum_{a_t} \nabla \pi_\theta(a_t|s_t) \\
&= r(s_k, a_k, s_{k+1}) \nabla \sum_{a_t} \pi_\theta(a_t|s_t) = r(s_k, a_k, s_{k+1}) \nabla 1 = 0
\end{aligned}$$

Policy Gradient Method

- Hence, we have

$$\begin{aligned}\nabla J(\theta) &= \mathrm{E}_{\pi_\theta}\left\{R(\tau)\sum_{t=0}^{\infty}\nabla\log\pi_\theta(a_t|s_t)\right\}\\ &= \mathrm{E}_{\pi_\theta}\left\{\sum_{k=0}^{\infty}\gamma^k r_{k+1}\sum_{t=0}^{\infty}\nabla\log\pi_\theta(a_t|s_t)\right\}\\ &= \mathrm{E}_{\pi_\theta}\left\{\sum_{t=0}^{\infty}\nabla\log\pi_\theta(a_t|s_t)\sum_{k=t}^{\infty}\gamma^k r_{k+1}\right\}\end{aligned}$$

- Policy gradient (reward-to-go version):

$$\nabla J(\theta) = \mathrm{E}_{\pi_\theta}\left\{\sum_{t=0}^{\infty}\nabla\log\pi_\theta(a_t|s_t)G_t\right\},\ \text{with } G_t = \sum_{k=t}^{\infty}\gamma^k r_{k+1}$$

Policy Gradient

Algorithm 5 Policy Gradient

Require: Initialize policy parameter θ
Require: Learning rate α, number of epoches R
  for loop R times **do**
    Initialize game and observe current state s_0
    Generate one episode $(s_0, a_0, r_1, s_1, a_1, r_2, \ldots, s_{T-1}, a_{T-1}, r_T)$ using policy $\pi_\theta(a \mid s)$
    for $t = 0, 1, \ldots, T-1$ **do**
      Compute return $G_t = \sum_{k=t}^{T-1} \gamma^{k-t} r_{k+1}$
      $\theta \leftarrow \theta + \alpha\, G_t \nabla_\theta \log \pi_\theta(a_t \mid s_t)$
    end for
  end for
  Output policy $\pi_\theta(a \mid s)$

Offline Policy Gradient

- Suppose $s_0 \sim \rho_0$ and we have a trajectory

$$\tau = (s_0, a_0, r_0, s_1, a_1, r_1, \ldots) \text{ generated from a behavior policy } \pi_b$$

- We still have

$$\begin{aligned}\nabla J(\theta) &= \mathrm{E}_{\pi_\theta}\left\{\sum_{t=0}^{\infty} \nabla \log \pi_\theta(a_t|s_t) G_t\right\} \\ &= \mathrm{E}_{\pi_b}\left\{\sum_{t=0}^{\infty}\left(\prod_{u=0}^{t} \frac{\pi_\theta(a_u|s_u)}{\pi_b(a_u|s_u)}\right) \nabla \log \pi_\theta(a_t|s_t) G_t\right\}\end{aligned}$$

  - Importance sampling:

$$\mathrm{E}(f(X)) = \int f(x) p_X(x) dx = \int f(x) \frac{p_X(x)}{p_Y(x)} p_Y(x) dx = \mathrm{E}\left(\frac{p_X(Y)}{p_Y(Y)} f(Y)\right)$$

Offline Policy Gradient

Algorithm 6 Offline Policy Gradient

Require: Initialize policy parameter θ
Require: Offline dataset $\mathcal{D} = \{\tau^{(i)}\}_{i=1}^{n}$ collected under behavior policy π_b
Require: Learning rate α, discount factor γ, number of epoches R

```
for loop R times do
    Sample one episode (s_0, a_0, r_1, s_1, a_1, r_2, ..., s_{T-1}, a_{T-1}, r_T) from D
    for t = 0, 1, ..., T-1 do
        Compute return G_t = sum_{k=t}^{T-1} γ^{k-t} r_{k+1}
        Compute importance weight W_t = prod_{u=0}^{t} π_θ(a_u | s_u) / π_b(a_u | s_u)
        θ ← θ + α W_t G_t ∇_θ log π_θ(a_t | s_t)
    end for
end for
Output policy π_θ(a | s)
```

Coding

- See instruction in class

More on Policy Gradient

- Policy gradient (reward-to-go version):

$$\nabla J(\theta) = \mathrm{E}_{\pi_\theta}\left\{\sum_{t=0}^{\infty} \nabla \log \pi_\theta(a_t|s_t)(G_t - b(s_t))\right\}$$

  - The equation holds for any functioin $b(s)$
  - Because $\mathrm{E}_{\pi_\theta}\left\{\nabla \log \pi_\theta(a_t|s_t) b(s_t)\right\} = 0$
- A good choice is $V^{\pi_\theta}(s_t)$
  - This motivates the Actor-critic methods
  - Actor to update θ
  - Critic to update $V_w(s) \approx V^{\pi_\theta}(s)$ using an additional parameter w

Offline Reinforcement Learning

- Offline Q-leanring:
  - Do not need to know the expression of π_b
- Offline policy gradient:
  - Need to know the expression of π_b
- Is this assumption feasible?